

The Weinberg Angle from Three Dimensions and Two Strands

The Two Axioms Behind $\sin^2\theta_W = 1/(3\sqrt{2})$

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Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Abstract

This paper isolates the two structural inputs — three spatial dimensions and two rope strands — that together produce the Weinberg-angle estimate $\sin^2\theta_W = 1/(3\sqrt{2})$, and asks how much of the value is forced by them. The '3' and the ' $\sqrt{2}$ ' each have a clear structural reading; what is not established is that these two inputs UNIQUELY force the specific combination $1/(3\sqrt{2})$ rather than merely being consistent with it. We state the estimate and this openness honestly. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants.

Scope of this paper

CLAIM: the estimate $1/(3\sqrt{2})$ rests on two structural inputs (3 dimensions, 2 strands) with clear readings.

NOT CLAIMED: that these inputs uniquely force $1/(3\sqrt{2})$. Uniqueness is Open; the ~2% gap persists. Status: DEMOTED 5 September 2026 (KOIDE-WEINBERG, EW-001) to a KEPT COINCIDENCE; this paper's own 'uniqueness is Open' is the first of three named choice points. See the Disposition section.

1. The Two Inputs

- Three spatial dimensions — supplying the factor 3.
- Two rope strands — supplying the $\sqrt{2}$.

Combined, these give $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$.

2. What Is Open

The honest gap is uniqueness: the two inputs are consistent with $1/(3\sqrt{2})$, but the paper does not establish that they FORCE it — that no other combination of the same inputs is admissible. Until

that is shown, and until the $\sim 1.94\%$ deviation from measurement is understood, the value remains a structurally-motivated conjecture rather than a derivation.

The honest status of this value. The predicted $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ differs from the measured 0.23122 by about 1.94%. It is close, and its form is suggestive, but it is NOT exact, and the test suite pins it explicitly as a 'soft external input' — the value that, when fed into the lepton-mass relation instead of the measured angle, breaks it. This is therefore a geometric conjecture of the right magnitude, not a precision derivation, and the $\sim 2\%$ gap is the known weak link of the electroweak sector.

Disposition (5 September 2026): demoted to a kept coincidence

Verdict. Commission KOIDE-WEINBERG (ITEM 6d) executed the derivation-or-demotion this paper's status line deferred. Under FND-088's protocol (require a dependency path from registered objects to the number, with every choice point listed, before looking at the target) the chain 'three dimensions supply the 3, two strands supply the $\sqrt{2}$ ' has three choice points on its face: (1) the combination rule, why $1/(3\sqrt{2})$ rather than $1/\sqrt{6}$, $1/3^2$, $\sqrt{2}/3$ or $1/(3+\sqrt{2})$, which the two-axioms paper itself records as 'uniqueness is Open'; (2) the quantity, why $\sin^2\theta_W$ carries the geometric number rather than $\sin\theta_W$, $\tan^2\theta_W$ or g'/g ; (3) the multiplicative reading of the two factors, for which the Hopf route supplies transparency but no forcing mechanism. A chain with choice points is a conjecture whose choices are now named. The registered verdict is EW-DEMOTE: EW-001 leaves Conjecture grade and is kept as a coincidence. Full record: analysis/KOIDE_WEINBERG_results.md.

The renormalization point, fixed before re-reading. A scale-free geometric number can be compared to exactly one renormalization point. The registered comparison, 0.23122 (the MS-bar value at M_Z), was locked before the comparison was re-read; against it the estimate is 1.94% off. The low- Q^2 effective angle (about 0.238) was displayed alongside and never adjudicated against: the estimate is 0.97% from it. A number within one to two percent of whichever point one picks is what a coincidence at that level looks like, and running or threshold effects are not invoked to close the gap.

What stands. The two structural readings (three dimensions, two strands) stand as readings; the 1.94% gap stands as measured; the test-suite pin stands. What is withdrawn is the word 'estimate' as a claim about the electroweak sector: the corpus holds no derived Weinberg angle. The companion lepton-mass relation (PM-001) was demoted in the same commission on independent grounds (its sensitivity to this angle is 71: a 1.94% change in the input moves the muon-to-electron mass ratio by a factor 7.8), which closes the coupled inconsistency the corpus carried between the two entries.

Status

$1/(3\sqrt{2})$ from two structural inputs: DEMOTED to a kept coincidence (EW-001, 5 September 2026); the two inputs stand as readings, not as a derivation.

Uniqueness of the combination — Open.

$\sim 1.94\%$ deviation — the known weak link.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.